

LGCP simulation pipeline (rLGCP)

$i = 1, \dots, N$

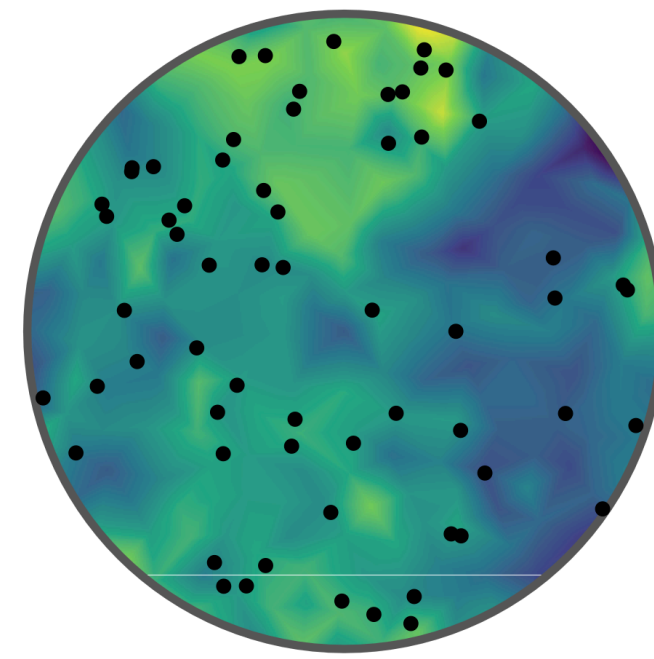
$N = n_{\text{train}} + n_{\text{val}} + n_{\text{test}}$

$\mu_i \sim \mathcal{U}(\mu_{\min}, \mu_{\max})$

$\sigma_i^2 \sim \mathcal{U}(\sigma_{\min}^2, \sigma_{\max}^2)$

$s_i \sim \mathcal{U}(s_{\min}, s_{\max})$

rLGCP



TARGETS

$\theta_i = (\mu_i, \sigma_i^2, s_i)$

INPUTS

$L_c(r) = \hat{L}(r) - r$
 N_i

$\mathbf{f}_i \in \mathbb{R}^8$

Quadrat-based (c_k)

$f_1 = \text{Var}(c_k)$
 $f_2 = \text{Var}(c_k) / \bar{c}$
 $f_3 = \max_k c_k / (\min_k c_k + 1)$

KDE-based ($\hat{\lambda}$)

$f_4 = \text{Var}(\hat{\lambda})$
 $f_5 = \text{skew}(\hat{\lambda})$
 $f_6 = \text{kurt}(\hat{\lambda})$
 $f_7 = H(\hat{\lambda})$
 $f_8 = \text{SD}(\hat{\lambda}) / \bar{\hat{\lambda}}$

Training

$i = 1, \dots, n_{\text{train}}$

Validation

$i = 1, \dots, n_{\text{val}}$

Test

$i = 1, \dots, n_{\text{test}}$

Dataset splits